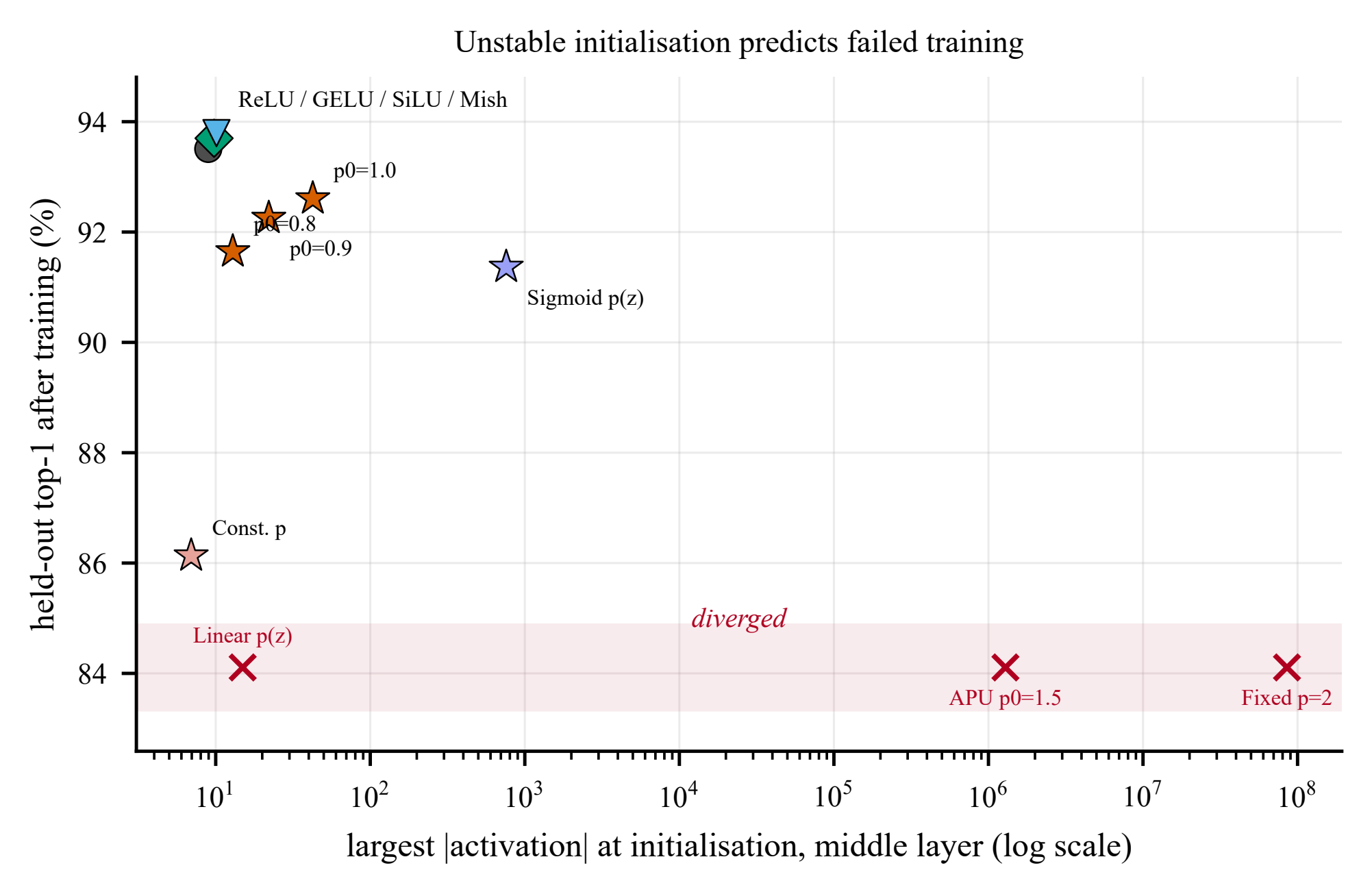
Unstable initialisation predicts failed training
ReLU / GELU / SiLU / Mish
p0=1.0
p0=0.8
p0=0.9
Sigmoid p(z)
Const. p
Linear p(z)
diverged
APU p0=1.5
Fixed p=2
held-out top-1 after training (%)
94
92
90
88
86
84
10^1
10^2
10^3
10^4
10^5
10^6
10^7
10^8
largest |activation| at initialisation, middle layer (log scale)